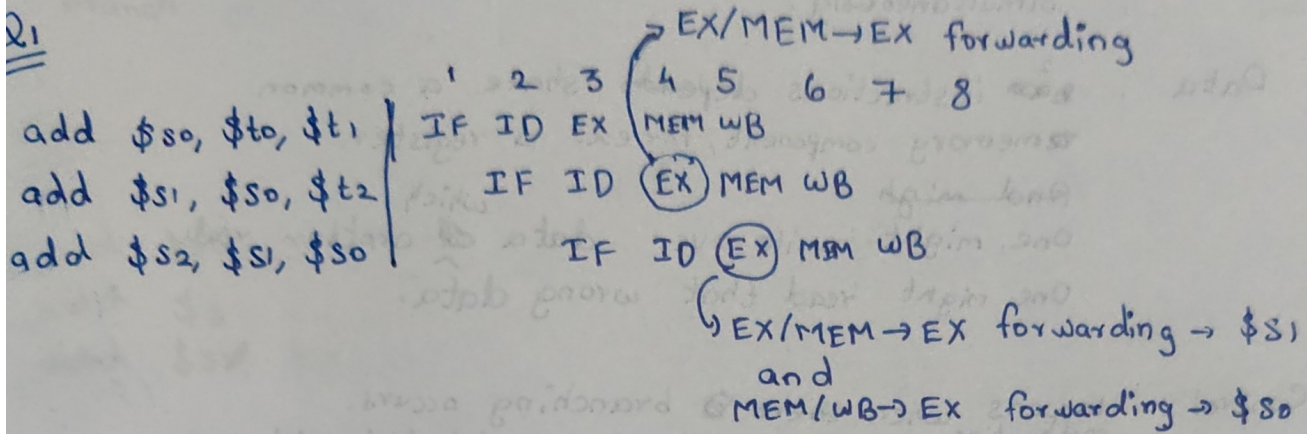


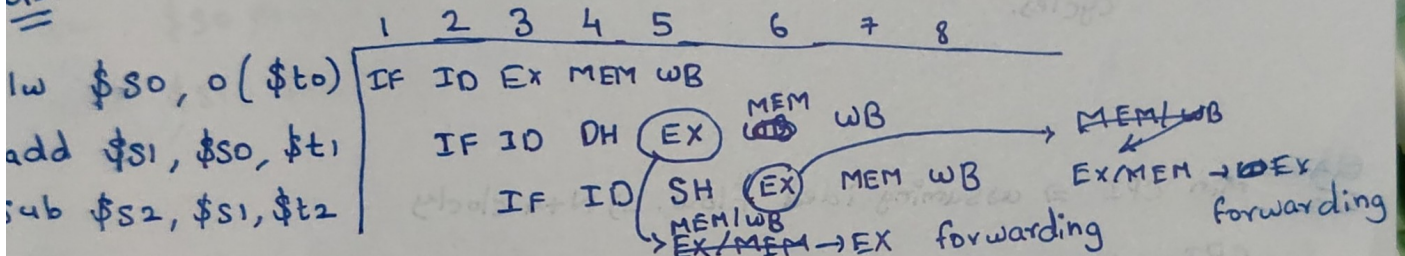
# SuperScalar week-1

## Part A:

Q1



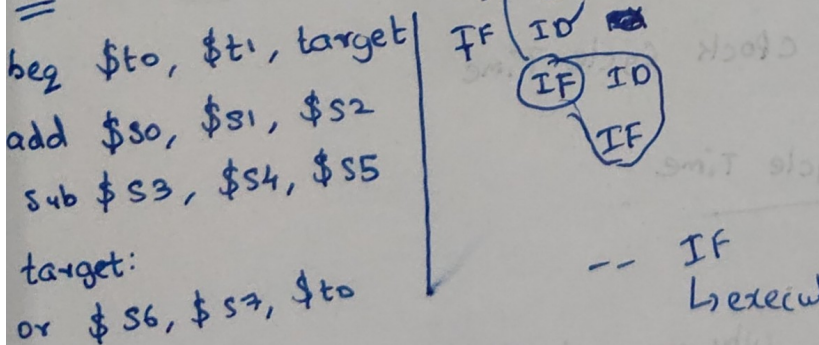
Q2



DH: Data Hazard due to since lw's output is only loaded from memory to MEM/WB register later.

SH: EX components are occupied.

Q3





## Part-B:

Q1 Structural: 2 instructions need same piece of hardware simultaneously.

Data: 2 instructions depend on share a common memory component, for ex. a register.  
And might one might write over data which another reads.  
One might read that wrong data.

Control: Happens when there is branching occurs.  
consecutive  
If instructions have already been fetched/decoded, they need to be flushed which consumes some cycles.

Q2 min CPI  $\Rightarrow$  assuming ideal flow.  $\Rightarrow N+4$  clocks

$CPI = \frac{N+4}{N}$  but.... if we consider initial 4 as just "startup" latency

then...

$$\boxed{CPI = \frac{N}{N} = 1}$$

Q3

Iron Law:

$$IPC =$$

$$CPU \text{ Time} = I.C \times CPI \times \text{Clock Cycle Time}$$

$$IPC = \frac{I.C \times \text{Clock Cycle Time}}{CPU \text{ Time}}$$

I mean I don't get why you absolutely want usage of "Iron Law" here, ~~for~~ each instructions takes one clock to execute, so it's just

$$\text{So } CPI=1 \Rightarrow IPC = \frac{1}{CPI} = 1.$$

Though, you say regardless of hardware budget which is misleading.



Q4

Add a comparator to compare PC output.

Branch penalty is 1 cycle on misprediction, on correct - none.

Q5

|                      |    |    |    |     |    |  |
|----------------------|----|----|----|-----|----|--|
| add \$s0, \$t0, \$t1 | IF | ID | EX | MEM | WB |  |
| add \$s1, \$t1, \$t2 | IF | ID | EX | MEM | WB |  |
| add \$s2, \$t0, \$s0 | IF | ID | EX | MEM | WB |  |

EX/MEM forwarding

lw \$s0, 10(\$t0)

add \$s1, \$s0, \$t0

Q6 write (to register file) happens only at WB stage.

In in-order execution W.B of later instruction cannot occur before W.B of preceding instructions, thus they are no problem.

In out-of-order execution, quite obviously W.B of later instructions can come ahead of preceding instructions thus it becomes relevant.